AAE 50700: Principles of Dynamics
Problem Set 8 Report

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(a): Given

Consider a system whose dynamics are described by the following Hamiltonian:

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2, \quad \boldsymbol{x} = \begin{bmatrix} q \\ p \end{bmatrix} \tag{1}$$

(a.1): Find

Derive the equations of motion using Hamilton's equations and confirm that it represents a harmonic oscillator.

Analysis/Conclusions:

Hamilton's equations for a one dimensional system with no external forces are written as:

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q} \tag{2}$$

So we solve the two partials to obtain the equations of motion:

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{\partial}{\partial p}\left(\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2\right) = \frac{\partial}{\partial p}\left(\frac{p^2}{2m}\right) + \frac{\partial}{\partial p}\left(\frac{1}{2}m\omega^2 q^2\right) = \frac{p}{m} + 0 = \frac{p}{m}$$

$$\dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial}{\partial q}\left(\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2\right) = -\frac{\partial}{\partial q}\left(\frac{p^2}{2m}\right) - \frac{\partial}{\partial q}\left(\frac{1}{2}m\omega^2 q^2\right) = -m\omega^2 q \tag{3}$$

Thus the equations of motion are:

$$\dot{q} = \frac{p}{m}, \quad \dot{p} = -m\omega^2 q \tag{4}$$

To prove this is a harmonic oscillator, we look at the defining equation for one:

$$\ddot{q} + \omega^2 q = 0 \tag{5}$$

where q is the single coordinate, and ω is the natural frequency.

To prove the EoM's are equivalent, we can rewrite $\dot{q}$ in a different form and take its time derivative:

$$\begin{gathered} \dot{q} = \frac{p}{m} \quad \Longrightarrow \quad \dot{q} - \frac{p}{m} = 0 \\ 0 = \frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{q} - \frac{p}{m}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{q}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{p}{m}\right) = \ddot{q} - \frac{\dot{p}}{m} \end{gathered} \tag{6}$$

Then substitute in the found expression for $\dot{p}$:

$$0 = \ddot{q} - \frac{\dot{p}}{m} = \ddot{q} - \frac{-m\omega^2 q}{m} = \ddot{q} + \omega^2 q \tag{7}$$

Which is the same form as the harmonic oscillator, proving that the equations of motion represent one.

(a.2): Find

For some constant $a \neq 0$, consider the coordinate-momentum scaling

$$q^* = aq, \quad p^* = \frac{p}{a} \tag{8}$$

Write the inverse transformation $\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{y})$, where $\boldsymbol{y} = [q^*, p^*]^\top$. Discuss whether this transformation is canonical or not using the symplectic condition:

$$F^\top J F = J, \quad F = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{y}}, \quad J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \tag{9}$$

Analysis/Conclusions:

To find the inverse transformation, we simply must invert the given equations for q^* and p^*:

$$q^* = aq \quad \Longrightarrow \quad q = \frac{q^*}{a}, \qquad p^* = \frac{p}{a} \quad \Longrightarrow \quad p = ap^* \tag{10}$$

Thus we write the inverse transformation as:

$$\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{y}) = \begin{bmatrix} a^{-1}q^* \\ ap^* \end{bmatrix} \tag{11}$$

Then to determine if the transformation is canonical, we solve the Jacobian F from the symplectic condition:

$$F = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{y}} = \begin{bmatrix} a^{-1}q^* \\ ap^* \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial q^*} & \frac{\partial}{\partial p^*} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial q^*}\left(a^{-1}q^*\right) & \frac{\partial}{\partial p^*}\left(a^{-1}q^*\right) \\ \frac{\partial}{\partial q^*}\left(ap^*\right) & \frac{\partial}{\partial p^*}\left(ap^*\right) \end{bmatrix} = \begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \tag{12}$$

Then we verify the condition:

$$\begin{aligned} F^\top JF &= \begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \\ &= \begin{bmatrix} a^{-1} \cdot 0 + 0 \cdot -1 & a^{-1} \cdot 1 + 0 \cdot 0 \\ 0 \cdot 0 + a \cdot -1 & 0 \cdot 1 + a \cdot 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \\ &= \begin{bmatrix} 0 & a^{-1} \\ -a & 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \\ &= \begin{bmatrix} 0 \cdot a^{-1} + a^{-1} \cdot 0 & 0 \cdot 0 + a^{-1} \cdot a \\ -a \cdot a^{-1} + 0 \cdot 0 & -a \cdot 0 + 0 \cdot a \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = J \end{aligned} \tag{13}$$

Thus $F^\top JF = J$, so the symplectic condition is satisfied. Since this condition is both necessary and sufficient, **the transformation is canonical for any** $a \neq 0$. We can see by observation that if $a = 0$, then F would be undefined because of the a^{-1} term. Thus the condition is only satisfied when $a \neq 0$.

(a.3): Find

Repeat the test for $q^* = aq$, $p^* = ap$. Determine the values of a for which this transformation is canonical (in the sense of the second condition covered in class).

Analysis/Conclusions:

Once again we invert the given equations for q^* and p^*:

$$q^* = aq \quad \Longrightarrow \quad q = a^{-1}q^*, \qquad p^* = ap \quad \Longrightarrow \quad p = a^{-1}p^* \tag{14}$$

Thus we write the inverse transformation as:

$$\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{y}) = \begin{bmatrix} a^{-1}q^* \\ a^{-1}p^* \end{bmatrix} \tag{15}$$

Then for the symplectic condition, we derive the Jacobian F:

$$F = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{y}} = \begin{bmatrix} a^{-1}q^* \\ a^{-1}p^* \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial q^*} & \frac{\partial}{\partial p^*} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial q^*}\left(a^{-1}q^*\right) & \frac{\partial}{\partial p^*}\left(a^{-1}q^*\right) \\ \frac{\partial}{\partial q^*}\left(a^{-1}p^*\right) & \frac{\partial}{\partial p^*}\left(a^{-1}p^*\right) \end{bmatrix} = \begin{bmatrix} a^{-1} & 0 \\ 0 & a^{-1} \end{bmatrix} \tag{16}$$

Then we verify the condition:

$$\begin{aligned} F^\top JF &= \begin{bmatrix} a^{-1} & 0 \\ 0 & a^{-1} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a^{-1} \end{bmatrix} \\ &= \begin{bmatrix} a^{-1} \cdot 0 + 0 \cdot -1 & a^{-1} \cdot 1 + 0 \cdot 0 \\ 0 \cdot 0 + a^{-1} \cdot -1 & 0 \cdot 1 + a^{-1} \cdot 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a^{-1} \end{bmatrix} \\ &= \begin{bmatrix} 0 & a^{-1} \\ -a^{-1} & 0 \end{bmatrix} \begin{bmatrix} a^{-1} & 0 \\ 0 & a^{-1} \end{bmatrix} \\ &= \begin{bmatrix} 0 \cdot a^{-1} + a^{-1} \cdot 0 & 0 \cdot 0 + a^{-1} \cdot a^{-1} \\ -a^{-1} \cdot a^{-1} + 0 \cdot 0 & -a^{-1} \cdot 0 + 0 \cdot a^{-1} \end{bmatrix} \\ &= \begin{bmatrix} 0 & a^{-2} \\ -a^{-2} & 0 \end{bmatrix} = a^{-2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = a^{-2} J \end{aligned} \tag{17}$$

Thus with the relation:

$$F^\top JF = a^{-2} J \tag{18}$$

$F^\top JF = J$ only when $a^{-2} = 1$. Solving for a:

$$\begin{aligned} a^{-2} &= 1 \\ \frac{1}{a^2} &= 1 \\ a^2 &= 1 \\ a^2 - 1 &= 0 \\ (a-1)(a+1) &= 0 \\ a &= \pm 1 \end{aligned} \tag{19}$$

Thus the transformation is canonical only when $a = \pm 1$ since that is when the symplectic condition is met.

(a.4): Find

Use the type-2 generating function

$$W_2(q, p^*) = aqp^* \tag{20}$$

to derive Eq. (8). For $a = \sqrt{m\omega}$, construct the transformed Hamiltonian $K(q^*, p^*)$ and derive its canonical equations. Discuss an advantage of the transformed form.

Analysis/Conclusions:

To derive Eq. (8) we can use the following relationship between the type 2

function W_2 and momentum p:

$$p = [\nabla_q W_2]_{q,p^*} = \frac{\partial}{\partial q}\left(aqp^*\right) = ap^* \frac{\partial q}{\partial q} = ap^* \tag{21}$$

Then we can invert p:

$$p = ap^* \quad \Longrightarrow \quad p^* = \frac{p}{a} \tag{22}$$

And then evaluate the relationship between the type 2 function W_2 and coordinate q^*:

$$q^* = [\nabla_{p^*} W_2]_{q,p^*} = \frac{\partial}{\partial p^*}\left(aqp^*\right) = aq \frac{\partial p^*}{\partial p^*} = aq \tag{23}$$

So, between the results of Eq. (22) and Eq. (23) the equations in Eq. (8) have been derived:

$$q^* = aq, \quad p^* = \frac{p}{a} \tag{24}$$

In part (a.2), it was already determined that the inverse transformation $\boldsymbol{f}(\boldsymbol{y})$ is canonical, which also implies it is invertable, and we can formulate the transformed Hamiltonian $K(q^*, p^*)$ by evaluating H at q^* and p^*. This requires the inverted equations for q^* and p^*:

$$q^* = aq \quad \Longrightarrow \quad q = \frac{q^*}{a}, \qquad p^* = \frac{p}{a} \quad \Longrightarrow \quad p = ap^* \tag{25}$$

Then we find K:

$$\begin{aligned} K(q^*, p^*) = [H]_{q^*,p^*} &= \left[\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2\right]_{q^*,p^*} = \frac{a^2 p^{*2}}{2m} + \frac{1}{2}m\omega^2 \frac{q^{*2}}{a^2} = \\ &= \frac{(\sqrt{m\omega})^2 p^{*2}}{2m} + \frac{m\omega^2 q^{*2}}{2(\sqrt{m\omega})^2} = \frac{m\omega p^{*2}}{2m} + \frac{m\omega^2 q^{*2}}{2m\omega} = \\ &= \frac{\omega p^{*2}}{2} + \frac{\omega q^{*2}}{2} = \frac{\omega}{2}\left(p^{*2} + q^{*2}\right) \end{aligned} \tag{26}$$

So the transformed Hamiltonian is:

$$K(q^*, p^*) = \frac{\omega}{2}\left(p^{*2} + q^{*2}\right) \tag{27}$$

Then we derive the canonical equations by the same process as (a.1):

$$\dot{q}^* = \frac{\partial K}{\partial p^*}, \quad \dot{p}^* = -\frac{\partial K}{\partial q^*} \tag{28}$$

Solving the two partials:

$$\dot{q}^* = \frac{\partial K}{\partial p^*} = \frac{\partial}{\partial p^*}\left(\frac{\omega}{2}\left(p^{*2} + q^{*2}\right)\right) = \frac{\omega}{2}\left(\frac{\partial p^{*2}}{\partial p^*} + \frac{\partial q^{*2}}{\partial p^*}\right) = \frac{\omega}{2}\left(2p^* + 0\right) = \omega p^*$$

$$\dot{p}^* = -\frac{\partial K}{\partial q^*} = -\frac{\partial}{\partial q^*}\left(\frac{\omega}{2}\left(p^{*2} + q^{*2}\right)\right) = -\frac{\omega}{2}\left(\frac{\partial p^{*2}}{\partial q^*} + \frac{\partial q^{*2}}{\partial q^*}\right) = -\frac{\omega}{2}\left(0 + 2q^*\right) = -\omega q^* \tag{29}$$

Thus the equations of motion are:

$$\dot{q}^* = \omega p^*, \quad \dot{p}^* = -\omega q^* \tag{30}$$

A major advantage of the transformed form is that the equations of motion depend only on the natural frequency ω, rather than on both m and ω as in the original form. The mass is absorbed into the scaling $a = \sqrt{m\omega}$, so the same pair of equations describes every harmonic oscillator, with m only needed when converting back to the physical coordinates.

ASK ABOUT THIS ONE

(b): Given

Let us consider a canonical transformation of the same space example as in PS7(b) to the spherical coordinates (r, ϕ, θ), where r, ϕ, θ are defined in class. Let the old canonical variables be $\boldsymbol{q} = [x, y, z]^\top$ and $\boldsymbol{p} = [p_x, p_y, p_z]^\top$. Define the new coordinates and corresponding momenta as follows:

$$\boldsymbol{q}^* = [r, \phi, \theta]^\top, \quad \boldsymbol{p}^* = [p_r^*, p_\phi^*, p_\theta^*]^\top \tag{31}$$

(b.1): Find

Express $\boldsymbol{q}$ as a function of $\boldsymbol{q}^*$ explicitly, and write it as $\boldsymbol{q} = \boldsymbol{f}(\boldsymbol{q}^*)$.